INCH-POUND
MIL-PRF-3098/105H
6 March 2013
SUPERSEDING
MIL-PRF-3098/105G
16 January 2006

PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR103/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 17 MHz to 61 MHz; third mechanical overtone; noncontrolled; series resonance.

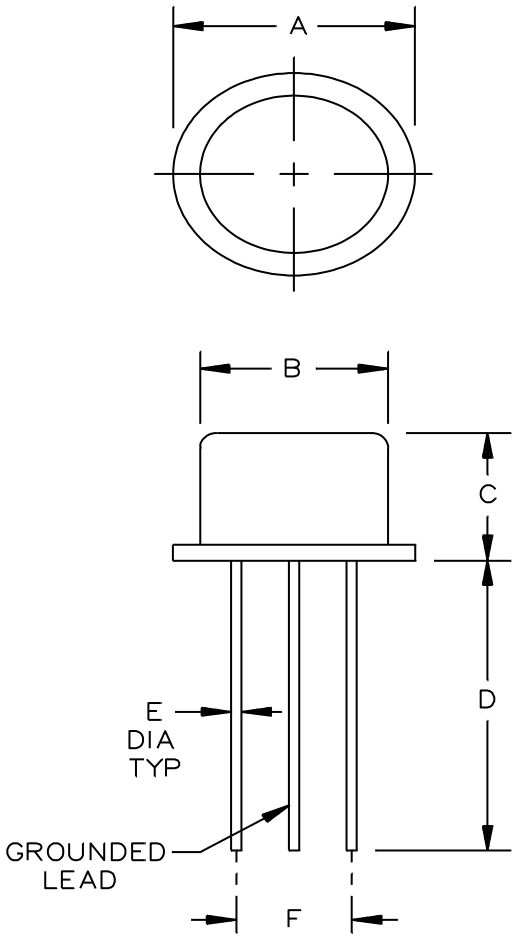


FIGURE 1. Crystal unit - CR103/U.

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Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.420	---	10.67
B	---	.326	---	8.28
C	---	.265	---	6.73
D	.600	---	15.24	---
E	.016	.020	.406	.508
F	.192	.208	4.88	5.28

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Type CR103/U, crystal unit - Continued.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 17 MHz to 61 MHz, inclusive.

Frequency tolerances: ±25 ppm.

Equivalent resistance: 50 ohms, maximum.

Mode of oscillation: Third overtone.

Operating temperature range (noncontrolled): -55°C to +105°C, inclusive.

Rated drive level: 1.0 mW, maximum.

Resonance: Series.

Capacitance, shunt: 7 pF, maximum.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent-resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent-resistance change permitted: ±10 percent.

Vibration: Method 204, MIL-STD-202, test condition B.

Frequency change permitted: ±5 ppm.

Equivalent-resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±2.5 ppm.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

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Navy - EC
Air Force - 85
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC, OS, SH
Air Force - 19, 99

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5955-2012-016)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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